

The Earnings Call Risk and Confidence Analyzer (ERCA)

Based on the unpublished working paper by:
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Abstract

The recent surge in zero-days-to-expiration (0DTE) options has damaged the retail traders significantly, leading to significant losses. These 0DTE options are increasingly difficult to predict due to the high volume of retail participation, leading to the options being highly volatile. High volatility leads to significant mispricing of the option compared to the social sentiment towards the option. This paper implements mathematical and machine learning models to successfully predict these significant differences in market price compared to social sentiment — referred to as divergence. Through a pipeline of mathematical processes and machine learning models, live social transcripts combined with official sentiment are used to predict these divergences and signal the model.

1 Introduction

This project implements the Earnings Call Risk and Confidence Analyzer (ERCA), a pipeline completely based on the in-progress, unpublished theoretical framework developed by Alejandro Herraiz Sen.

The modern options market has shifted dramatically toward short-duration strategies that implement the idea of zero days to expiration options. These zero-days-to-expiration (0DTE) options are options contracts that expire the same day that they were traded, which is perceived as potential for generating quick profits. This quick form of options trading is looked at by many as the optimal way to make a quick profit, as it can almost represent modern day gambling in options contracts. The general results of 0DTE options are either making a quick profit, or generally losing a significant amount, if not all of the money invested in the 0DTE option. This enticing method of profit attracts many retail traders, with more than 75% of their trades in the S&P 500 involving 0DTE contracts. These numbers would lead to believing that

the 0DTE options are profitable to retail investors, or at minimum lead to a better average net yield, however the opposite proves to be true. Retail investors lost an average of \$241,000 per day for the entire sample prior to the expansion of 0DTE trading, and now is estimated as an average loss of \$350,000 in 0DTE trading. The transaction costs, volatility, and unforgiving formatting works against the retail traders, making it difficult to make a profit from 0DTE options consistently.

To avoid these systematic losses that retail traders suffer from, the Earnings Call Risk and Confidence Analyzer (ERCA) proposes a mathematically driven solution to detect the mispricing of 0DTE options. Rather than attempting to predict the future price movements or direction of the option, which is inconsistent, the ERCA identifies when the price is too high and looks to profit from the imminent drop in volatility. Through a process of identifying divergences between retail sentiment and market pricing, a divergence signal is sent when a rapid social sentiment relative to the implied volatility is identified. When this divergence signal surpasses a threshold, it implies that options are overpriced and will experience an implied volatility crush. This can be used to avoid the volatility trap, and be leveraged as a 0DTE options strategy.

2 Literature Review

The mathematical foundation of our Earnings Call Risk and Confidence Analyzer is based on our group member Alejandro Herraiz Sen’s unpublished research paper on *The Earnings Call Risk and Confidence Analyzer (ERCA): A Stochastic Framework for Multi-Modal Sentiment Divergence, Latent Profile Dynamics, and 0DTE Options Mispricing*. Alejandro’s work showcases a complete theoretical framework that influences our ERCA. He first identifies a phenomenon called the “Volatility Trap” in which retail participation in 0DTE options inflames im-

plied volatility before earnings events, which cause a predictable IV crush that harms retail traders. To model the Volatility Trap, Herraiz Sen proposes a three-filtration decomposition separating the market, official, and social information channels. The official channel is modeled from a compound Poisson process scored through FinBERT, with the social media dynamics being captured by a Hawkes self-exciting process with an $\mathcal{O}(1)$ recursive update with real-time processing. It also includes latent investor psychology that is formalized through Latent Profile Analysis across $K = 8$ archetypes. The three channels are unified into a sentiment-coupled jump-diffusion SDE, which is then transformed into a risk-neutral pricing measure using Girsanov’s theorem. The resulting volatility coefficient makes closed-form solutions impossible and requires numerical PDE methods for option valuation. The divergence between the social sentiment velocity and the market pricing is captured through the Z_{short} divergence indicator, which fires a signal when it surpasses a certain threshold. This indicates that an option is overpriced and an IV crush is likely to occur. Herraiz Sen validates his framework on 500 real S&P 500 earnings events, finding a Spearman $\rho = 0.4773$ between the Z_{short} signal and EPS surprise magnitude, with top-quartile events generating signals $2.51\times$ stronger than the rest. A Deep Q-Network is used to act on the divergence signal, dynamically selecting the best-performing model from Neural CDE, Multi-Transformer, Bi-Transformer, and SVR based on the current state of the market, while risk is controlled through a fractional Kelly allocation that caps exposure at 25% of capital.

There are several other supporting works that influenced our project pipeline. Nandi’s survey on options and volatility shows that the Black-Scholes assumption of constant volatility is unreliable in practice. Nandi also mentions that stochastic models like Heston’s mean-reverting variance process better capture real options market behavior. Our pipeline addresses this directly by coupling volatility to live sentiment streams rather than just relying on a fixed volatility specification. Araci introduces FinBERT, which is a domain-adapted transformer model that outperformed general-purpose NLP tools in financial sentiment tasks. FinBERT also suggests future integration with options data and volatility signaling, which our pipeline directly ended up using. A Berkeley study on earnings call transcript analysis shows that semantic content from transcripts carries market signals, but finds that it struggles when predicting price direction. This study pushes us towards focusing on volatility mispricing detection rather than directional forecasting.

3 Data

The data used for this project has different types of data for analysis and cleaning. Real time social media posts were needed to get live updates through a social sentiment

scoring process, as well as the live option pricing through Yahoo Finance. Each script handles a different layer of the pipeline. SEC filings for 8-K filings. Yahoo Finance for market and options data. Reddit and StockTwits for public investor discussion, and VADER sentiment scoring for turning text into simple bullish, bearish, or neutral signals. From 8-K filings, to social media posts, and then to the actual market data which is all crucial for our project as they collect different types of information that can be used to explain the market.

The SEC EDGAR data contains the recent 8-K filings from the SEC website. These are official company documents that are important to investors and often result in high market movement. These contain reliable company information that can be connected to stock and options movement. Instead of relying only on social media or news, this data adds an official source of information to use.

Our team used Yahoo Finance to collect the stock data using yfinance. It can pull current stock information, historical prices, earnings dates, options chains, implied volatility, and trading volume. Which are all extremely useful and needed for predicting the implied volatility. This provides the actual market data needed for analysis. The team used this data to understand how the market reacted around earnings or major company events.

The Reddit and StockTwits data was created by collecting posts from public social media communities. It searches sub-Reddit’s like r/wallstreetbets, r/options, and r/posts. StockTwits was used to pull messages around the stock ticker. This is very important for our project because our team can see what social media is saying about a stock in real time. These posts can be used to explain market hype, fear, bullish options, or bearish options. This gives us a way to compare social media with options movement.

The sentiment scoring uses VADER sentiment analysis that can score something as positive, negative, or neutral. This can be used on the Reddit, StockTwits, or 8-K filing data that was collected. This is impactful for our project as sentiment scoring turns our text straight into a number. This makes it very easy to then use this as a feature in machine learning.

4 Machine Learning Pipeline

The framework for ERCA is a multi-stage pipeline that transforms social and official business data into signals detecting the mispricing in 0DTE options. The pipeline inputs data through mathematical models prior to using reinforcement learning to find an optimal policy for leveraging machine learning models. The pipeline consists of four phases consisting of, sentiment representation, stochastic state construction, divergence detection, and model selection and execution.

4.1 Sentiment Representation

Raw social transcript and market data are input into various sentiment scoring models — social data uses FinBERT and market data uses VADER — to represent text as a score. These sentiment scores (S_i) are assigned a value, $S_i \in [-1, 1]$ associated with a positive or negative correlation to the assigned business. This establishes social and market sentiment, which will be used for future calculations.

4.2 Stochastic State Construction

The sentiment scores are used as inputs to further expand on their relations to the market environment. Social intensity is modeled using a Hawkes process, to capture social transcript frequencies and clustering. The intensity function is representative of the beginning of the connection between the sentiment scores and the further connected signals that make up the end goal. Furthering on the intensity function the sentiment coupled jump-diffusion defined the volatility as a function of both sentiment and intensity. The volatility is the main feature for firing signals to further attempt to understand market behavior.

4.3 Divergence Detector

defined by the difference in the market value and social sentiment — signals to the modeling. The divergence indicates that there is an imbalance in pricing rather than predicting what is going to happen to the price. It represents the possibility of sentiment score falsely thriving above what the market price assumes it should be at. When the divergence (Z_{short}) surpasses a predetermined threshold (Γ) a signal is fired indicating the potential mispricing. This phase determines whether to continue the process into the models or whether to terminate without a signal.

4.4 DQN Ensemble & Execution

Rather than relying on a machine learning model to predict the price and direction of the option — which is statistically possible to consistently achieve — ERCA deploys a Deep Q-Network (DQN) to select from a set of models based on the market environment. The models chosen were Neural CDE, Multi-Transformer, Bi-Transformer, and SVR which all optimize different market settings. At each time step, the DQN observes which model performs the best throughout the environment and selects the model to use, creating an optimum Q-Learning policy. The models observe features such as volatility, sentiment intensity, and divergence to make decisions with an error decision of RMSE. By learning an optimum policy of model selection over a traditional machine learning approach will improve performance of the volatile 0DTE market environments.

5 Modeling

Note: The mathematical models, differential equations, and volatility feedback loops detailed in this section are directly derived from the unpublished theoretical framework currently under active development by Alejandro Herraiz Sen.

5.1 The Coupled SDE

The underlying asset dynamics of the ERCA framework are constructed to fuse traditional quantitative finance with real-time sentiment streams. Under the physical measure $\mathbb{P}$, the underlying asset price follows a specialized stochastic differential equation. This jump-diffusion process is mathematically defined as:

$$\frac{dS_t}{S_{t-}} = \mu(t, \mathcal{S}_{\text{off}})dt + \sigma(t, \tilde{\mathcal{S}}_{\text{soc}})dW_t + (e^{J_t} - 1)dN_t^{\text{off}} \quad (1)$$

This equation successfully couples standard price action with both official corporate disclosures and social media trading behavior.

The drift component of this equation is directly modulated by verified information releases. Specifically, the drift is formulated as:

$$\mu(t, \mathcal{S}_{\text{off}}) = \mu_0 + \eta \mathcal{S}_{\text{off}}(t) \quad (2)$$

This ensures that the expected localized price return path continuously updates as official sentiment state variables, parsed from tools like SEC EDGAR and corporate transcripts, enter the market filtration.

To capture the erratic variance induced by retail speculation, the diffusion term is explicitly driven by social sentiment rather than a constant or purely price-derived stochastic variable. The term $\sigma(t, \tilde{\mathcal{S}}_{\text{soc}})$ functions as the profile-weighted social sentiment volatility coupling. Consequently, as social media exuberance scales, it proportionately injects variance into the asset’s modeled price path.

Finally, the equation accounts for the sudden, discontinuous price shocks that are standard during earnings events. The jump timing is governed by N_t^{off} , which controls disclosure jump timing. When a jump occurs, its size J_t is normally distributed according to:

$$J_t \sim \mathcal{N}(\mu_J(\mathcal{S}_{\text{off}}(t^-)), \sigma_J^2) \quad (3)$$

This effectively ties the magnitude of discontinuous price corrections directly to the official sentiment state just prior to the jump.

5.2 Volatility Feedback Specification

The volatility feedback loop represents a major methodological extension, moving the pipeline away from generic sentiment analysis to a structured, predictive variance model. The specification explicitly extends the scalar approach to incorporate profile-weighted sentiment into the

diffusion coupling. This relationship is formally defined as:

$$\sigma(t, \tilde{\mathcal{S}}_{\text{soc}}) = \sigma_{\text{base}} + \ln(1 + \lambda_{\text{soc}}(t))\tilde{\mathcal{S}}_{\text{soc}}(t) \quad (4)$$

This theoretical relationship is deeply rooted in contemporary market microstructure research. The specification directly generalizes the empirical finding of Eaton et al. (2026), which dictates that implied volatility falls when retail demand is removed. By tying the Hawkes intensity of social posts, $\lambda_{\text{soc}}(t)$, to the volatility state, the model provides a quantitative structure to this demand-side pressure.

One of the most critical mechanical features of this specification is its ability to mathematically simulate the “IV crush” that routinely devastates retail option buyers. Following an earnings event, as the social intensity $\lambda_{\text{soc}}(t)$ decays back to its baseline μ_{soc} , the overall volatility $\sigma(t)$ deterministically collapses toward σ_{base} . This guarantees that the framework accurately prices the rapid deflation of options premiums post-event.

Furthermore, the integration of latent profiles protects the framework from misinterpreting dangerous market exuberance as a simple positive signal. The profile membership directly enters the specification through the variable $\tilde{\mathcal{S}}_{\text{soc}}$. Crucially, this profile-weighted state can be negative even when the raw, scalar textual sentiment is positive, effectively capturing the “irrational exuberance” mechanism documented by Eckhaus (2026).

5.3 Risk-Neutral Measure and Girsanov Transformation

To utilize the coupled SDE for the practical valuation of 0DTE options, the framework must transition from modeling the physical price path to establishing a pricing measure. This is achieved through a change of measure from the physical measure $\mathbb{P}$ to the risk-neutral measure $\mathbb{Q}$. By applying the Girsanov Transformation, the model adjusts the Brownian motion to accurately account for the market price of risk driven by these sentiment streams.

The specific transformation alters the stochastic process by netting out the sentiment-driven drift against the risk-free rate r_f , scaled by the sentiment-coupled volatility. The adjusted Brownian motion under the risk-neutral measure is explicitly defined as:

$$dW_t^{\mathbb{Q}} = dW_t + \left(\frac{\mu(t, \mathcal{S}_{\text{off}}) - r_f}{\sigma(t, \tilde{\mathcal{S}}_{\text{soc}})} \right) dt \quad (5)$$

This fundamental adjustment ensures the market modeled is free of arbitrage.

Applying this transformation yields the definitive sentiment-adjusted pricing equation used for derivative valuation. Under the risk-neutral measure $\mathbb{Q}$, the asset’s dynamics become:

$$\frac{dS_t}{S_{t-}} = r_f dt + \sigma(t, \tilde{\mathcal{S}}_{\text{soc}}) dW_t^{\mathbb{Q}} + (e^{J_t} - 1) dN_t^{*, \text{off}} \quad (6)$$

While the drift is successfully reduced to the risk-free rate r_f , the complex social sentiment volatility and the official disclosure jumps remain critical drivers of the option’s value.

Because of this sophisticated dependence on real-time data, valuing options using this equation carries significant computational requirements. The time-varying, profile-conditioned volatility term $\sigma(t, \tilde{\mathcal{S}}_{\text{soc}})$ enters the final pricing PDE as a non-constant coefficient. As a result, closed-form solutions are impossible, requiring the implementation of numerical PDE methods such as Crank-Nicolson or ADI finite difference schemes to execute option valuation.

6 Training

Unlike a typical model where one machine learning model is being trained, ERCA is training a model selector that chooses the best model based on the market environment. At each time step t , the Deep Q-Network (DQN) observes the market factors including, volatility, intensity, divergence condition, and sentiment features to select a prediction model. The DQN is trained on a 21-day split of realized volatility, and tested across data that is unseen in a 70/30 split. After observing the outcome of the selected model, the DQN is updated using Q-learning, and repeats over many iterations to learn a policy of the optimal model by market environment. The temporal difference loss (TD-loss) — the measurement of how wrong the DQN’s prediction is — is recorded across iterations to update the policy. Over iterations the TD-loss decreases as the policy improves learning what model is optimal for different conditions.

The results from the DQN training are recorded through RMSE and the improvement to the models prediction before and after training for the optimal policy. Across the four models used — Neural CDE, Multi-Transformer, Bi-Transformer, and SVR — SVR performed the best before the DQN training took place. The Multi-Transformer and Bi-Transformer performed similarly, and the Neural CDE performed significantly worse before training.

Noticeable improvements are present in the modeling after the DQN took place, lowering the error of almost every model (See Table 1). DQN improves the performance of the models by making the correct decision on which model to use in market settings opposed to teaching the models specifically. The overall ensemble learning method improved by approximately 5.7 % error recorded by RMSE, which provides a significant improvement in results.

7 Visualized Results

The mathematical modeling serves as the foundation of the ERCA framework. However, the ability to effectively

Model	Before(%)	After(%)	Δ
Neural CDE	20.32	6.30	-14.01
Multi-Transformer	6.02	6.02	0.00
Bi-Transformer	6.03	6.03	0.00
SVR	3.66	2.77	-0.89
Ensemble	9.01	3.34	-5.67

Table 1: Per-model RMSE improvement

visualize the mathematical findings allows for improved reasoning for the final execution of the pipeline. Visualization of intermediate properties of ERCA is critical in translating the mathematical process to interpretable results that can be leveraged in future work. These visualizations make the model more transparent and visually possible to validate in real time prior to execution. This leads the visualization to be derived as more than a presentation tool, but a component of the pipeline that assists in interpretability and execution.

7.1 Sentiment

The social sentiment score is derived from the public opinion on the company from various active social media websites. The sentiment radar displays whether the public is currently speaking in a bullish or bearish tone about the company. This radar alone cannot be a standalone indicator, as the public can falsely persuade others based on public trends that are actively taking place. A positive announcement could be indicative of a high social sentiment score, but it could very quickly come back to neutral and have no significant impact towards the ODTE option. The recent posts are displayed through the page for verification purposes, and determining the real reaction of the public. FinBERT and other sentiment text scorers are not perfect, and can mislabel possible bullish or bearish activity, but the recent results prove validation for the present results. The news sentiment is similarly derived from the selected companies official statements and documents. Recent business sentiment scores scraped from Yahoo Finance are visually stored in a bar plot of the net sentiment score. These net business sentiment scores combined with the multi-channel information releases spanning official filings (SEC 8-K) are key drivers of the divergence signal that we are searching for to identify mispricing. The social and business sentiments are driving factors towards the execution of the ERCA, and the visualization is important in understanding why future decisions are made.

7.2 ERCA Pipeline

ERCA is a four phase pipeline that follows through the phases of, data ingestion, stochastic state variables, divergence detection, and DQN Ensemble and execution. The live data streams captured in the sentiment tab are pro-

cessed through language models and statistical methods to determine behavioral signals. The resulting signals are transformed into stochastic state variables that are used to model the volatility and intensity behavior. Through analyzing the stochastic variables, ERCA computes a divergence signal when the market pricing deviates from the sentiment and volatility. This signal is the driving factor towards the ensemble and execution phase, when ultimately decisions are made through modeling decisions.

7.2.1 Data Ingestion

Phase A consists of providing a visualization towards the calculated sentiment scores that will be used for future calculations and interpretation. The official sentiment provides the official FinBERT sentiment scores for the previous ten official transcripts. The social channel sentiment — captured through FinBERT and an LPA profiler — is provided to capture the social sentiment among active traders. These are alternative displays of the aforementioned sentiment tab, but are an integral component of the ERCA pipeline. Without the display of the sentiment scoring the interpretation of the modeling loses interpretability and becomes significantly more theoretical.

7.2.2 Stochastic State Variables

Creating the stochastic state variables is where the process of turning raw data into mathematical signals begins. Now that the sentiment scores have been collected, it is now converted into dynamic variables that drive the function of the model. The market filtration is analyzed through the examination of both social intensity ($\lambda_{soc}(t)$) and the adjusted social sentiment ($\tilde{S}_{soc}(c)$). The social intensity — based mathematically on the Hawkes process — measures the posting frequency as well as the clustering of the social activity. The value of $\lambda_{soc}(t)$ can be interpreted by the user, as a high $\lambda_{soc}(t)$ indicates significant posting activity relative to time, and a low value indicates the inverse. This is coupled with the adjusted social sentiment through an LPA profiler to identify the signal’s strength. Individually these variables have little correlation to the signal, but when both variables are significantly sized, the signal strength is strong. The connection between sentiment and volatility is also present through the Sentiment-Coupled Jump-Diffusion graph. The model-implied volatility ($\sigma(t)$) indicates that volatility increases when both social sentiment and intensity are large, which is an important visualization for explaining why implied volatility is typically falsely inflated. This intensity is demonstrative of the fundamental basis of the ERCA modeling, providing a quantitative description for identifying ODTE option mispricing.

7.2.3 Divergence Detection

The divergence detection is the process that fires signals when there is a high probability that the option’s price is

mispriced due to excessive social retail sentiment. The divergence signal ($Z_{\text{short}}(t)$) gets a measurement of how far social sentiment has diverged from the listed market price. A signal is fired when $Z_{\text{short}} > \Gamma$ -Threshold, which indicates a divergence in the market price, telling the pipeline to proceed to the modeling and execution. The visualization of $Z_{\text{short}}(t)$ over a period of time, not only identifies the divergence in price, but also demonstrates the duration and size of the divergence. A divergence significantly above the threshold and a divergence slightly above the threshold are indistinguishable if only the signal was fired when it reached over the threshold. The presented graph portrays the confidence of the divergence that tends to lead to a larger market correction. A larger divergence — represented when the threshold is surpassed — tends to represent a larger IV crush, and looks to have more profitable confidence. While the divergence detection does not definitively represent profit, it does give a higher confidence that the signal sent for divergence is real.

7.2.4 DQN Ensemble & Execution

The work up to this point in the pipeline works towards detecting the mispricing of 0DTE options, and now the model is addressing how it can be acted on. Rather than acting on just a singular model, ERCA uses multiple — Neural CDE, Multi-Transformer, Bi-Transformer, and SVR — to optimize across different market conditions. Using a Deep Q-Network (DQN), the model learns which model gives the best predictions depending on the state of the market. This allows the model to adapt to the active market conditions, optimizing the overall modeling. The DQN displays which model to trust at t — represented by days to expiration — in relation to the realised implied volatility. Alongside the technical aspect of model optimization, the results from the modeling is also displayed. The signal and execution panel displays the τ^* condition on when the model will signal, the signal of whether to fire or hold out, and the Fractional Kelly Sizing — how much money to put into a trade based on the quality of the signal. The Fraction Kelly Sizing will determine the sizing of how much capital to invest with at a given time t up to 25% of capital. This final optimized value is indicative of how the user should respond to the resulting pipeline.

7.3 Model Exploration

The visualization illustrates the core components of the ERCA pipeline and their live dynamic interaction. The Hawkes self-exciting process demonstrates to the user how social activity spreads and how it clusters. Spikes in $\lambda(t)$ represent the sharp increase in social posts, demonstrating how a spike in posts can result in several more grouped spikes. Social posts can iteratively feed off itself by showing up on a media feed and resulting in more posts related to the idea. The Latent Profile Analysis

(π_k) identifies trends in sentiment behavior, attempting to display any irrational behavior. The market may be dominated by a certain type of investor, i.e. Informed Bull, which can cause irrational behavior in the sentiment data. The $Z_{\text{short}}(t)$ divergence indicator is reinforced here, displaying the importance of not forcing trades, but waiting for a strong divergence in sentiment and market data before firing a signal. If the model detects significantly mispriced market value, a signal will fire to the user indicating a signal to go in. However the model does not suggest you go all in with your capital from the previously mentioned Fractional Kelly Sizing, which is reinforced again, with the drawdown — the measurement of how much has been lost from the highest point — being tracked in the relative graph. The drawdown will capture the risk alongside the Fractional Kelly Sizing to optimize the modeling and determine when to deploy capital.

8 Results

In the historical backtest, the full ERCA pipeline was replayed for a year of real price data. Social events were synthesized via a Hawkes process seeded from actual weekly volatility — larger price moves generate more posting activity. Sentiment scores were correlated with price direction with introduced noise, matching the empirical social media behavior around earnings events.

Through this backtest, the model fired a signal identifying a market divergence a total of 29 weeks, approximately once every other week. These signals only fire when the Z_{short} is greater than the threshold (0.01), which is not a common occurrence. Timing is one of the main issues with 0DTE options, as it is very difficult to properly identify the best times to enter due to the increased volatility of the option. The threshold is set to validate that the model will correctly identify divergence, and not get caught up in flash trends. Varying the threshold to a larger margin decreases the frequency at which the signal is fired, but does not consistently increase the average annual return.

Metric	Value
Total weeks	52
Signal weeks	29
Signal rate	55.8%
Avg return (signal)	0.13%
Avg return (quiet)	1.29%
Spearman ρ	-0.0591
Signal threshold (Γ)	0.01

Table 2: Signal accuracy values of backtest period

Of the 29 signal periods fired during the 52 week period, the model had 55% of fired signals yielding a 1.29% profit margin (see Table 2). The model correctly fired 16 times, but does not maximize its yield due to the lack of divergence from the model. The model shows an overall

accuracy of 52% across the window (see Figure 1). While profits are left in the open, ERCA’s design is not looking to make the optimal decision every time — the volatility spread across 0DTE options makes this feat statistically impossible — but to optimize the times when it is sure to make capital. In a market that yields large negative losses annually to retail traders, a net positive increase of 1.29% of capital is significant in relation to the average mean capital.

	Signal (Trade)	No Signal
Profitable Outcome	16	12
Unprofitable Outcome	13	11

Figure 1: Confusion matrix of 52-week backtest period

9 Conclusion

The rapid increase in 0DTE option trading, and therefore the net losses, has introduced a problem due to the unpredictability of the options. Through the implementation of the Earnings Call Risk and Confidence Analyze — Based on the unpublished working paper by Alejandro Herraiz Sen — an increase in identifying the divergence between market value and social sentiment was established using a mathematical pipeline. Through using these divergence signals in a Deep Q-Network reinforcement learning model, an average return of 1.3% was achieved. The integration of the mathematical pipeline and modeling of ERCA successfully developed a predictive model for 0DTE options through divergence signals.

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